E. Test

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Sometimes it is hard to prepare tests for programming problems. Now Bob is preparing tests to new problem about strings — input data to his problem is one string. Bob has 3 wrong solutions to this problem. The first gives the wrong answer if the input data contains the substring s_1, the second enters an infinite loop if the input data contains the substring s_2, and the third requires too much memory if the input data contains the substring s_3. Bob wants these solutions to fail single test. What is the minimal length of test, which couldn't be passed by all three Bob's solutions?

Input

There are exactly 3 lines in the input data. The i-th line contains string s_i. All the strings are non-empty, consists of lowercase Latin letters, the length of each string doesn't exceed 10^5.

Output

Output one number — what is minimal length of the string, containing s_1, s_2 and s_3 as substrings.

Examples

input Copy

```
ab
bc
cd
```

output Copy

```
4
```

input Copy

```
abacaba
abaaba
x
```

output Copy

```
11
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 25 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372250080	Apr/23/2026 16:38	Accepted
372249776	Apr/23/2026 16:35	Accepted
372249349	Apr/23/2026 16:32	Accepted

372248763	Apr/23/2026 16:27	Runtime error on test 40
372248475	Apr/23/2026 16:25	Compilation error
372247871	Apr/23/2026 16:20	Runtime error on test 40
372247622	Apr/23/2026 16:18	Runtime error on test 40
372247311	Apr/23/2026 16:15	Time limit exceeded on test 12
372247150	Apr/23/2026 16:14	Compilation error
372246907	Apr/23/2026 16:12	Wrong answer on test 7

→ Problem tags

hashing strings *2200

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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